Review 2

Project Title - CyberTrace: Cyber Threat Intelligence and Attack Relationship Explorer

Review 2 evaluates the implemented Neo4j database, CRUD functionality, and advanced NoSQL features. The repository meets every Review 2 scoring criterion. A fresh verification run on 25 September 2026 confirmed 159 nodes, 1,116 relationships, complete CRUD, ten working investigation queries, and 90 valid multi-hop paths.

Review 2 Rubric Mapping

Criterion	Marks	Implemented evidence	Status
Database implementation	2	7 labels, 6 relationship types, 159 nodes, 1,116 relationships, sample data and provenance	Complete
CRUD operations	3	Node and relationship create, read, update, and delete through Neo4j-backed APIs and the Manage Data UI	Complete
Advanced NoSQL features	5	10 Cypher investigations, multi-hop traversal, indexes, constraints, transactions, aggregations, and graph visualization	Complete

1 Database Implementation

1.1 Implementation Overview

CyberTrace uses Neo4j Community 5.26.31 as its operational database. The backend uses Flask and the official Neo4j Python driver. The browser interface calls a JSON API for dashboard metrics, threat search, entity details, graph exploration, CRUD, and investigation queries. Database access stays in the backend so credentials are never sent to the browser.

Layer	Repository location	Responsibility
Data preparation	backend/data and scripts	Imports MITRE ATT&CK, generates controlled synthetic records, validates provenance, and prepares deterministic JSON artifacts
Database	backend/db.py and backend/schema.py	Manages Neo4j sessions, transactions, constraints, and indexes
Application service	backend/services.py	Implements entity, relationship, dashboard, search, and graph operations
API	backend/api.py	Exposes validated HTTP endpoints with stable error responses
User interface	frontend	Provides dashboard, search, graph explorer, manage data, and investigation views

1.2 Implemented Graph Schema

The implementation expands the five-label proposal from Review 1 into seven labels. Vulnerability and Organization support exploit and impact scenarios, while the original ThreatActor, Malware, Technique, IP, and Domain labels remain central.

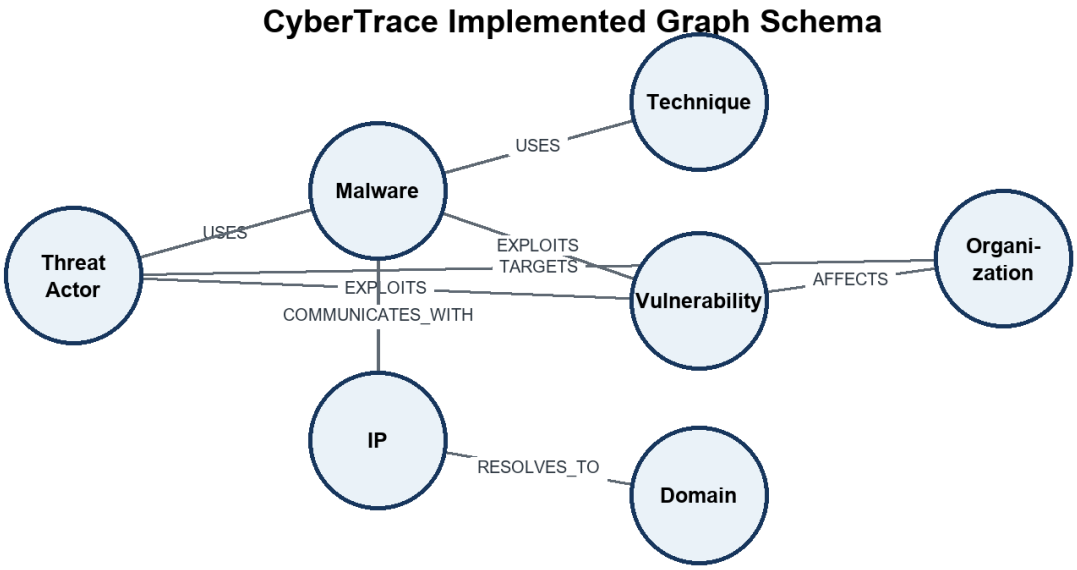


Figure 1 Implemented CyberTrace graph schema

1.3 Node and Relationship Design

Node label	Stable identifier	Count	Purpose
ThreatActor	actor_id	25	Named adversary groups from MITRE ATT&CK
Malware	malware_id	40	Malware, tools, and software used by actors
Technique	technique_id	40	ATT&CK techniques associated with actors or software
Vulnerability	cve_id	12	Controlled synthetic vulnerability scenarios
IP	ip_id	15	Documentation-safe synthetic infrastructure addresses
Domain	domain_id	15	Example synthetic domain infrastructure
Organization	organization_id	12	Controlled synthetic target organizations

Relationship type	Implemented pattern	Purpose
USES	ThreatActor or Malware to Malware or Technique	Documents ATT&CK usage associations
TARGETS	ThreatActor to Organization	Represents a controlled target scenario

Relationship type	Implemented pattern	Purpose
EXPLOITS	ThreatActor or Malware to Vulnerability	Represents a controlled exploit scenario
COMMUNICATES_WITH	Malware to IP	Connects software to synthetic infrastructure
RESOLVES_TO	IP to Domain	Connects IP and domain infrastructure
AFFECTS	Vulnerability to Organization	Provides the sixth distinct relationship type required by the rubric

1.4 Dataset Design and Provenance

The database combines a traceable real subset with a controlled synthetic layer. Real records come from the official MITRE ATT&CK Enterprise 19.2 STIX 2.1 bundle. The importer verifies the pinned SHA-256 checksum before transformation. Every real record stores its ATT&CK or STIX identifier, source URL, dataset version, retrieval time, and is_synthetic=false.

Synthetic records provide safe infrastructure, vulnerability, and target relationships that are not asserted by the selected ATT&CK subset. They use deterministic seed 302, reserved documentation address ranges, example domains, generator metadata, and is_synthetic=true. This separation prevents demonstration data from being presented as real threat intelligence.

Dataset component	Nodes	Relationships	Identification
MITRE ATT&CK Enterprise 19.2	105	1,038	source fields and is_synthetic=false
Controlled synthetic fixture	54	78	Synthetic source, generator version, seed 302, and is_synthetic=true
Combined Neo4j database	159	1,116	Live totals verified by scripts.verify_review2

1.5 Constraints Indexes and Repeatable Loading

Seven uniqueness constraints protect the stable identifier for each node label. Six explicit search indexes cover actor, malware, technique, IP, domain, and organization lookup fields. Neo4j also reports seven constraint-backed indexes. The loader uses MERGE with stable identifiers, so rerunning the import does not create duplicates.

```
CREATE CONSTRAINT threatactor_actor_id_unique IF NOT EXISTS
FOR (n:ThreatActor) REQUIRE n.actor_id IS UNIQUE

CREATE INDEX malware_name_index IF NOT EXISTS
FOR (n:Malware) ON (n.name)
```

2 CRUD Operations

2.1 Node CRUD

The API supports create, read, update, and delete operations for all seven allowlisted labels. The backend chooses labels and identifier fields from fixed metadata rather than accepting raw Cypher fragments. Property values remain parameters. Manual records are forced to synthetic provenance so a user cannot create a record that appears to come from MITRE ATT&CK.

Operation	Endpoint	Implementation behavior
Create	POST /api/entities	Validates the label and required stable identifier, assigns manual synthetic provenance, and creates the node
Read	GET /api/entities/type/id	Returns all stored properties for one entity
Update	PUT /api/entities/type/id	Updates only mutable properties defined for that entity type
Delete	DELETE /api/entities/type/id	Uses DETACH DELETE so the node and attached relationships are removed safely
Search	GET /api/entities/search	Searches each supported label through an allowlisted field and returns provenance

2.2 Relationship CRUD

Relationship creation accepts only implemented source-label, relationship-type, and target-label patterns. Relationship values use parameters, and each created relationship receives a stable relationship_id and synthetic provenance. DELETE /api/relationships/id removes the selected relationship without deleting either endpoint node.

2.3 Demonstrated CRUD Workflow

The Manage Data page provides the reviewer workflow. The demonstration creates a malware node named ReviewDemoMalware, reads it by ID, changes its properties, confirms that search returns the update, deletes it, and verifies that the record no longer exists. The live script completed all six checks during final verification.

- Create synthetic malware - PASS
- Read malware by stable identifier - PASS
- Update mutable malware properties - PASS
- Search for the updated value - PASS
- Delete the malware and attached relationships - PASS
- Verify that the malware was deleted - PASS

```
python -m scripts.verify_crud
```

3 Advanced NoSQL Features

3.1 Parameterized Cypher Investigations

CyberTrace implements ten named investigations in backend/investigations.py. Each query has a defined purpose, accepted parameter, and maximum result limit. Final verification executed every query against the live database.

Investigation	Question answered	Verified rows
actor-malware	Which malware is used by a named threat actor	2
malware-techniques	Which techniques are used by named malware	1
actor-targets	Which organizations are targeted by a named actor	1
malware-vulnerabilities	Which vulnerabilities are exploited by named malware	1
malware-infrastructure	Which IP and domain infrastructure is connected to named malware	1
actor-investigation	Which actor to malware to IP to domain paths exist	2
actor-techniques	Which techniques are directly associated with a named actor	10
actor-vulnerabilities	Which synthetic vulnerabilities are linked to a named actor	1
vulnerability-organizations	Which synthetic organizations are affected by a vulnerability	1
most-connected	Which entities have the highest graph degree	10

3.2 Multi Hop Graph Traversal

The actor-investigation query follows a four-node path from ThreatActor through Malware and IP to Domain. This traversal is the central graph use case identified in Review 1. The live database contains 90 complete paths of this form.

```
MATCH (a:ThreatActor)-[:USES]->(m:Malware)
      -[:COMMUNICATES_WITH]->(i:IP)
      -[:RESOLVES_TO]->(d:Domain)
WHERE a.name = $value
RETURN properties(a), properties(m), properties(i), properties(d)
LIMIT $limit
```

3.3 Aggregation Transactions and Safe Query Construction

The dashboard aggregates nodes by label, relationships by type, and real versus synthetic provenance. The most-connected investigation uses count(r) to rank entities by graph degree without calling the result a risk score. Database operations use driver-managed read and write transactions. Labels and relationship types come from strict allowlists; user values remain parameters.

3.4 Graph Explorer and Responsive Interface

The Graph Explorer requests a bounded neighborhood for a selected entity and renders the result as an interactive SVG graph. Nodes, relationship labels, and properties are available without an external visualization service. The interface also includes responsive layouts and explicit empty, validation, success, connection, and error states.

3.5 Current Scope of Graph Analytics

Degree ranking and multi-hop traversal are implemented for Review 2. Bounded shortest-path analysis and a second graph analytics feature are scheduled for Review 3. They are not counted as completed work in this report.

4 Intermediate Demonstration and Results

4.1 Demonstration Sequence

- Open the dashboard and confirm 159 nodes, 1,116 relationships, label totals, and the provenance split.
- Search for APT28 and inspect the MITRE ATT&CK source identifiers and is_synthetic=false value.
- Open Graph Explorer for ThreatActor G0007 and inspect connected malware, techniques, and relationship labels.
- Use Manage Data to create, read, update, and delete the ReviewDemoMalware record.
- Run the actor-investigation query to display an actor to malware to IP to domain path.
- Run the one-command Review 2 verification and confirm the final readiness message.

4.2 Final Verification Results

Verification area	Observed result	Outcome
Generated artifacts	159 nodes, 1,116 relationships, 7 labels, 6 types, real and synthetic provenance	PASS
Live Neo4j	7 constraints, 6 explicit indexes, 7 constraint-backed indexes, 90 multi-hop paths	PASS
CRUD workflow	Create, read, update, search, delete, and deletion verification	PASS
Cypher investigations	All 10 named queries returned valid results	PASS
Automated tests	20 tests passed in 0.97 seconds	PASS

The complete verification command ended with CYBERTRACE REVIEW 2 READY on 25 September 2026.

```
python -m scripts.verify_review2
```

5 Repository Organization and Reproduction

5.1 Main Evidence Locations

Location	Evidence
backend/schema.py	Seven uniqueness constraints and six explicit search indexes
backend/services.py	Node CRUD, relationship CRUD, search, dashboard, and graph exploration
backend/investigations.py	Ten parameterized Cypher investigations
backend/data	Real importer, synthetic generator, validation, and loader logic
frontend	Dashboard, search, graph explorer, manage data, and investigation interface
scripts/verify_review2.py	One-command database, CRUD, and investigation verification
tests	Twenty automated tests for configuration, schema, import, validation, loader, API, and investigations
docs/review2.md	Reviewer-facing criterion mapping and demonstration order

5.2 Reproduction Commands

```
powershell -ExecutionPolicy Bypass -File scripts/start_neo4j.ps1
python -m scripts.load_neo4j
python -m backend.app
python -m scripts.verify_review2
python -m pytest -q
```

Database credentials are read from the local .env file. The repository ignores the credential file, the local Neo4j runtime, and the 53.8 MB raw source bundle. Processed data, the synthetic fixture, source manifest, and checksum remain in the project for reproducibility.

6 Conclusion

CyberTrace satisfies the Review 2 database implementation, CRUD, and advanced NoSQL criteria. The implementation exceeds the minimum 50-record expectation with 159 graph nodes and records 1,116 relationships. It also meets the broader Neo4j structural minimums for labels, relationship types, node count, ten meaningful Cypher queries, interactive graph visualization, and responsive UI. Review 3 work remains limited to the declared future scope: shortest-path analysis, a second graph analytics feature, final usability polish, and final presentation materials.

References

1. The MITRE Corporation. MITRE ATT&CK Enterprise data, release 19.2, STIX 2.1 bundle, released 5 August 2026. https://github.com/mitre-attack/attack-stix-data/releases/tag/v19.2
2. MITRE ATT&CK. Terms of Use. https://attack.mitre.org/resources/legal-and-branding/terms-of-use/
3. Neo4j. Cypher Manual and Python Driver Documentation. https://neo4j.com/docs/